

# Cosmic Expansion as Surface-Area Particle Creation in a Dirac–Kuramoto Substrate

A Coherence-Based Reframing of the FRW Universe and the Cosmological Constant

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## Abstract

The standard Friedmann–Robertson–Walker (FRW) cosmology treats expansion as a geometric property of spacetime sourced by a cosmological constant  $\Lambda$  inserted by hand into Einstein’s field equations.  $\Lambda$  originated as Einstein’s 1917 patch to recover a *static* universe — a Newtonian-era prejudice — and was later resurrected as “dark energy” to fit accelerating expansion. In both incarnations,  $\Lambda$  enters the framework axiomatically rather than emergently.

This paper proposes a reframing in which cosmic expansion is *not* a property of an underlying expanding spacetime, but instead the dynamical growth of a finite *populated region* embedded in an infinite, static quantum-field substrate. Expansion proceeds via real-particle creation from vacuum fluctuations in regions of low temperature and low gravitational density — the regime in which the Dirac–Kuramoto (DK) decoherence

rates  $\Gamma_{\text{env}}(T)$  and  $\Gamma_{\text{grav}}(M)$  both vanish. Because creation occurs preferentially at the boundary of the populated region, the rate of expansion is proportional to the **surface area** of the universe rather than its volume, naturally producing a Hubble rate that scales as  $H \propto 1/R$  and an effective cosmological term  $\Lambda_{\text{eff}} \propto 1/R^2$  — matching the observed  $\Lambda \sim 1/L_H^2$  without postulation.

The framework predicts three observational signatures distinct from  $\Lambda$ CDM: (1) the local Hubble rate should anti-correlate with local matter density, yielding the observed  $H_0$  tension between CMB and local-distance-ladder measurements; (2) cosmic voids should expand faster than virialized structures by a measurable margin; (3) the surface-area scaling implies an “emergent FRW metric” inside the populated region that preserves Tolman  $(1+z)^4$  dimming and CMB blackbody structure, while the underlying substrate remains Minkowski. The proposal is presented as a working-paper sketch, not a fully derived theory; the load-bearing assumption — that excitations of the bulk-coherence order parameter  $\Phi_{\text{bulk}}$  inherit equation of state  $w = -1$  through their Goldstone-mode character — is identified as the next required derivation.

## 1. Introduction

### 1.1 The framework problem with $\Lambda$

Modern cosmology rests on the Friedmann–Robertson–Walker (FRW) metric, an exact solution of Einstein’s field equations under the *Cosmological Principle* — the assumption that the universe is homogeneous and isotropic at large scales. The Friedmann equations governing the time evolution of the FRW scale factor  $a(t)$  take the form:

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3} \quad (\text{Friedmann I})$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3} \quad (\text{Friedmann II})$$

where  $\rho$  is the total energy density,  $p$  its pressure,  $k \in \{-1, 0, +1\}$  the spatial-curvature index, and  $\Lambda$  the cosmological constant. The framework is extraordinarily successful: it reproduces Hubble’s law (recession velocity proportional to distance), predicts the cosmic microwave background (CMB) and its near-perfect blackbody spectrum, accounts for primordial nucleosynthesis, and — with the addition of a positive  $\Lambda$  — fits the accelerating expansion measured by Type Ia supernovae from 1998 onward.

The framework’s success does not, however, dissolve the conceptual problem with  $\Lambda$  itself.  $\Lambda$  enters Einstein’s equations as a free parameter chosen to match observation. Its origin is unspecified by the theory. Three distinct unsolved problems attach to it:

1. **The cosmological-constant problem proper.** Quantum field theory predicts a vacuum energy density of order  $\rho_{\text{vac}} \sim M_{\text{Pl}}^4$ , roughly 120 orders of magnitude larger than the observed  $\rho_{\Lambda} \sim 10^{-47} \text{ GeV}^4$ . Why these two quantities differ so radically is the largest known discrepancy between theory and observation in all of physics.
2. **The coincidence problem.** Why is  $\rho_{\Lambda}$  comparable in magnitude to  $\rho_{\text{matter}}$  in the current cosmic epoch, given that the two have entirely different scaling with  $a(t)$ ?
3. **The  $H_0$  tension.** Independent measurements of the current expansion rate disagree at  $\sim 5\sigma$ . CMB-anchored measurements (Planck) give  $H_0 \approx 67.4 \text{ km/s/Mpc}$ ; local distance-ladder measurements (SH0ES) give  $H_0 \approx 73.0 \text{ km/s/Mpc}$ .  $\Lambda$ CDM with a single, uniform  $\Lambda$  cannot accommodate both.

Each of these problems suggests that  $\Lambda$  is *not* the right primitive — that expansion ought to emerge from some deeper dynamics, not be postulated.

## 1.2 The Dirac–Kuramoto framework: what is already available

The DK framework [companion paper, PAPER\_UNIFIED] proposes that quantum measurement consists of two stages: (Stage 1) pair-synchronization of a perturbed partner with a macroscopic *bulk* characterized by an order-parameter phase  $\Phi_{\text{bulk}}$ , followed by (Stage 2) irreversible re-equilibration of the partner to the bulk via two parallel rate channels,

$$\Gamma_{\text{Stage 2}}^{\text{total}} = \Gamma_{\text{env}}(T, J(\omega)) + \Gamma_{\text{grav}}(M, \Delta z) \quad (3.5')$$

where  $\Gamma_{\text{env}}$  is the environmental (thermal + bath-mediated) relaxation rate and  $\Gamma_{\text{grav}}$  is the gravitational self-relocking rate of the bulk. Stage-2 events are the rare bath-coupling moments that produce classical, irreversible measurement records.

Both rates vanish in the limit of low temperature and low gravitational density:  $\Gamma_{\text{env}} \rightarrow J(\omega)$  as  $T \rightarrow 0$  (the spontaneous-emission floor), and  $\Gamma_{\text{grav}} \rightarrow 0$  as the local matter density goes to zero. The DK framework therefore already specifies the conditions under which bulk-coherence-mediated decoherence ceases to act.

The proposal of this paper is that *exactly those conditions* — low T and low gravity — define the regime in which the cosmological substrate converts vacuum fluctuations into real particles, and that this conversion is the physical source of cosmic expansion.

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## 2. A Brief History of FRW Cosmology

### 2.1 Einstein’s static universe (1917)

Einstein applied his newly completed field equations of general relativity to the universe as a whole and found that no static solution existed for a homogeneous matter distribution under attractive gravity alone — the universe would either expand or contract. Believing on philosophical grounds that the universe must be eternal and static, Einstein modified his equations by adding a term  $\Lambda g_{\mu\nu}$ ,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu},$$

where  $\Lambda$  was tuned to exactly balance gravitational attraction at cosmic scales. The resulting “Einstein static universe” was a closed 3-sphere of fixed radius.

### 2.2 Friedmann and Lemaître (1922, 1927)

The Russian mathematician Alexander Friedmann (1922), and independently the Belgian priest-physicist Georges Lemaître (1927), solved Einstein’s equations under the assumptions of homogeneity and isotropy *without* demanding stasis. They found that the universe must in general expand or contract, with the dynamics governed by Friedmann I (above). Lemaître went further, connecting the expanding solutions to observational redshift data and proposing what would later be called the Big Bang. Einstein famously rejected Lemaître’s physics — “your calculations are correct, but your physics is abominable” — before reversing himself after Hubble’s 1929 observations.

### 2.3 Hubble’s law (1929)

Edwin Hubble’s observation that galactic recession velocity is proportional to distance,

$$v = H_0 d \quad (\text{Hubble’s law})$$

established expansion empirically. Einstein subsequently called his introduction of  $\Lambda$  his “greatest blunder.”

## 2.4 Robertson and Walker (1935–36); FRW metric

Howard Robertson and Arthur Walker independently proved that the most general metric consistent with homogeneity and isotropy takes the form

$$ds^2 = -c^2 dt^2 + a(t)^2 \left[ \frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right] \quad (\text{FRW metric})$$

where  $a(t)$  is the scale factor encoding all time evolution and  $k \in \{-1, 0, +1\}$  labels open, flat, and closed spatial geometries respectively. Observations of CMB power-spectrum peaks indicate  $k = 0$  to within a fraction of a percent.

## 2.5 The accelerating universe (1998) and $\Lambda$ 's return

Type Ia supernova surveys (Riess et al. 1998; Perlmutter et al. 1999) showed that the universe's expansion is *accelerating*. Within FRW, only a component with sufficiently negative pressure — equation of state  $w < -1/3$  — can drive acceleration. The simplest such component is a positive cosmological constant ( $w = -1$ ), and “ $\Lambda$ CDM” became the standard cosmological model. The dark-energy term thereby resurrected was mathematically Einstein's original  $\Lambda$  but with no static-universe motivation — purely an empirical fit.

## 2.6 The standing question

In its modern form,  $\Lambda$  is a single number tuned to one observation (acceleration) that fits a wide range of others (CMB, large-scale structure, BAO) — but with the three open problems noted in §1.1. The framework proposed below leaves the FRW machinery intact as an *effective* description of expansion inside the populated region, while replacing the fundamental ontology with one in which  $\Lambda$  is emergent.

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# 3. The Substrate–Population Distinction

## 3.1 The standard fusion

In standard FRW cosmology, two distinct ontological commitments are fused:

- **Spacetime is the universe.** The FRW metric describes the entirety of physical space; there is no “outside” of the universe in the framework.
- **The metric itself expands.** The scale factor  $a(t)$  describes expansion of space, not just motion of matter through space.

These commitments are taken together as a single package, and the cosmological-constant problem inherits both:  $\Lambda$  is both the energy density of spacetime itself and the agent of its expansion.

## 3.2 The DK ontology

The DK framework suggests a separation:

- **The substrate:** an infinite, eternal quantum-field structure (the vacuum and its fluctuations). The substrate has no privileged scale, no boundary, and is not itself expanding. Its metric, in the absence of coherent excitation, is Minkowski.
- **The populated region:** the finite, growing subset of the substrate in which real-particle excitations exist and are bulk-coherent in the DK sense. This is what we call “the universe.” It has a boundary, a scale, and a dynamical evolution.

Cosmic expansion, on this view, is **not** the metric of the substrate stretching. It is the *boundary of the populated region advancing into the substrate* as vacuum fluctuations cross a coherence threshold and become real excitations.

### 3.3 Why the substrate metric remains Minkowski

The substrate is the unexcited vacuum of quantum field theory. QFT vacuum states are Lorentz invariant — they look the same in every inertial frame. The unique Lorentz-invariant metric is Minkowski. So the substrate, by its Lorentz symmetry alone, has no expansion to speak of: there is no scale factor in Minkowski space because there is no scale at all.

What we observe as “expansion” must therefore be a property of the populated region’s internal structure, not of the substrate.

### 3.4 The effective FRW metric inside the populated region

Within the populated region,  $\Phi_{\text{bulk}}$  is established and matter-energy is present at non-zero density. The DK framework already predicts that bulk coherence determines an effective metric on the coherent region: the order-parameter phase  $\Phi_{\text{bulk}}$  couples to local proper-time evolution and to gravitational pair-locking. Internally, observers measure an effective metric that — by symmetry, since the populated region is homogeneous and isotropic on average — takes FRW form. The scale factor  $a(t)$  that appears in this effective metric is determined by the dynamics of the populated region itself, not by any property of the substrate.

This is structurally analogous to *analogue gravity* in condensed-matter systems: phonons propagating in a Bose–Einstein condensate experience an effective Lorentzian metric set by the condensate density and flow [Unruh 1981, Barceló–Liberati–Visser 2005]. The analogue metric is real for its internal excitations even though the underlying atoms live in flat lab space. Likewise, the effective FRW metric is real for matter inside the cosmological populated region, even though the underlying QFT substrate is Minkowski.

## 4. The Creation Mechanism: Low $\Gamma$ as the Threshold

### 4.1 Vacuum fluctuations and the virtual–real distinction

Quantum field theory predicts continuous vacuum fluctuations: pairs of virtual particles appearing and annihilating on timescales bounded by the energy–time uncertainty relation  $\Delta E \cdot \Delta t \sim \hbar$ . Under ordinary conditions, virtual pairs annihilate before they can be detected or do measurable work.

Several known mechanisms convert virtual pairs into *real* particles by preventing the annihilation:

| Mechanism                    | Trigger               | Conversion source                      |
|------------------------------|-----------------------|----------------------------------------|
| Schwinger pair production    | Strong electric field | Field energy                           |
| Hawking radiation            | Event horizon         | Horizon geometry                       |
| Unruh effect                 | Acceleration          | Kinetic energy of accelerated observer |
| Parker cosmological creation | Expanding metric      | Metric work                            |
| Dynamical Casimir effect     | Moving boundary       | Boundary kinetic energy                |

Each of these mechanisms identifies a *separation* of the virtual pair before they recombine. The energy that becomes the real particles’ rest-mass-equivalent is supplied by the external agent (field, horizon, acceleration, expansion, boundary motion).

### 4.2 The DK threshold: low $\Gamma$ as a fifth conversion mechanism

The DK framework specifies a different conversion mechanism. In ordinary matter, virtual pairs are re-absorbed because the bulk’s coherence dynamics — specifically  $\Gamma_{\text{env}}(T)$  and  $\Gamma_{\text{grav}}(M, \Delta z)$  — quickly damp any fluctuation back into the bulk-coherent state before it can become a measurement-relevant real excitation. Bulk coherence acts as an active *suppressor* of vacuum fluctuations.

In regions of low temperature and low gravitational density, both rates vanish:

$$\Gamma_{\text{env}}(T) \rightarrow J(\omega) \cdot [1 + 2n_{\text{th}}(\omega, T)] \xrightarrow{T \rightarrow 0} J(\omega)$$

$$\Gamma_{\text{grav}}(M, \Delta z) \rightarrow 0 \quad \text{as} \quad M_{\text{local}} \rightarrow 0$$

When both rates fall below the inverse lifetime of a virtual fluctuation, the fluctuation is no longer re-absorbed in time. It survives long enough to become a real excitation of the field — a real particle. The suppression mechanism has failed.

**The proposal:** *cosmic expansion is sourced by the failure of bulk-coherence suppression in low- $\Gamma$  regions of the substrate.* The same vacuum fluctuations that occur ubiquitously become real only where  $\Gamma_{\text{env}}(T)$  and  $\Gamma_{\text{grav}}(M)$  cannot damp them in time. These regions are: (i) cosmic voids, (ii) the cold dark sky between galaxies and clusters, and (iii) most critically, the *boundary* of the populated region itself, where the density of pre-existing matter is by definition lowest.

### 4.3 Where the energy comes from

The energy budget for new particles must come from somewhere. The DK framework points to a natural source: the **incompleteness of bulk coherence at horizon scales**.  $\Phi_{\text{bulk}}$  cannot complete its sync beyond the causal horizon; the residual coherence deficit is an energy reservoir analogous to the spontaneous-emission floor  $J(\omega)$  in the Stage-2 environmental term. When a virtual pair near the horizon fails to be reabsorbed, it draws on this reservoir to upgrade to real-particle status. The expansion of the populated region thereby converts horizon-incomplete-coherence energy into new real-particle excitations.

### 4.4 The equation-of-state problem

A standing technical problem remains: *real particles created from the vacuum carry positive energy density, which gravitates, producing contraction rather than expansion.* For the mechanism to drive expansion, the created excitations must inherit an equation of state  $w \approx -1$  (the cosmological-constant equation of state) rather than  $w = 0$  (cold matter) or  $w = 1/3$  (radiation).

The DK framework provides a natural candidate: the created excitations are not localized matter quanta but **collective phase modes (Goldstone modes) of the broken-U(1) order parameter  $\Phi_{\text{bulk}}$** . Goldstone modes of a condensate are massless by Goldstone’s theorem, propagate relativistically in the absence of a preferred frame, and — in Lorentz-covariant formulations — carry vacuum-like equation of state  $w = -1$ . The created excitations are thereby identified with *expansion-driving* quanta, not gravitating matter.

This identification is the load-bearing assumption of the proposal and the next step in its derivation. A complete treatment must show, from the broken-U(1) structure of cosmological  $\Phi_{\text{bulk}}$ , that its Goldstone modes carry  $w = -1$  to the required precision and that their creation rate at low  $\Gamma$  reproduces the observed expansion history.

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## 5. The Surface-Area Scaling

### 5.1 The geometric observation

Real-particle creation occurs preferentially where  $\Gamma_{\text{env}}$  and  $\Gamma_{\text{grav}}$  are smallest. Within the populated region, both rates are non-zero (matter is present, and the CMB sets a non-zero temperature floor). At the *boundary* of the populated region, however, both rates vanish — by definition, this is where the matter density of the populated region goes to zero.

The boundary is approximately a 2-sphere of radius  $R(t)$  (set by the extent of the populated region in cosmic time), with surface area

$$A(t) = 4\pi R(t)^2.$$

The rate of real-particle creation at the boundary is proportional to the area of the boundary times a per-area creation rate  $\sigma_c$  set by the strength of vacuum fluctuations and the relevant coupling:

$$\frac{dN}{dt} = \sigma_c \cdot A(t) = 4\pi\sigma_c R(t)^2 \quad (5.1)$$

## 5.2 The Hubble rate as a function of R

If the rate of *volume* growth of the populated region is proportional to its surface area times a characteristic “creation-front speed”  $v_c$ ,

$$\frac{dV}{dt} = v_c \cdot A(t),$$

then for a spherical region  $V = \frac{4}{3}\pi R^3$ , we have  $\frac{dV}{dt} = 4\pi R^2 \dot{R}$ , giving

$$\dot{R} = v_c \quad (5.2)$$

— the *boundary advances at constant speed*  $v_c$ . Identifying the scale factor  $a(t) \propto R(t)$ , the effective Hubble rate is

$$H = \frac{\dot{a}}{a} = \frac{\dot{R}}{R} = \frac{v_c}{R(t)} \quad (5.3)$$

The Hubble rate naturally **decreases as**  $1/R$  as the universe grows. At fixed  $v_c$ , an older, larger universe expands more slowly in fractional terms even as its boundary advances at constant absolute speed.

## 5.3 The effective cosmological constant

Comparing (5.3) with the  $\Lambda$ -dominated form of Friedmann I,

$$H^2 = \frac{\Lambda c^2}{3} \implies \Lambda_{\text{eff}} = \frac{3H^2}{c^2} = \frac{3v_c^2}{c^2 R^2} \quad (5.4)$$

so the effective cosmological constant scales as  $1/R^2$ , matching the observed scaling  $\Lambda \sim 1/L_H^2$  where  $L_H \approx c/H_0$  is the Hubble length.

This is, at the order-of-magnitude level, the correct value for the observed  $\Lambda$  — without postulating  $\Lambda$  at all. The framework produces a dynamically generated cosmological term whose magnitude is set by the size of the populated region.

## 5.4 Connection to the holographic principle

The surface-area scaling has a deep precedent in physics: the **holographic principle** (Bekenstein, 't Hooft, Susskind, Verlinde), which holds that the maximum entropy enclosable in a region is bounded not by its volume but by its surface area in Planck units,

$$S \leq \frac{A}{4\ell_P^2}.$$

Verlinde’s *entropic gravity* and related proposals extend the holographic principle to suggest that gravity itself, and possibly the cosmological constant, arise from entropy gradients at boundaries.

The DK proposal is in the same family: expansion is sourced at the horizon, with rate proportional to horizon area, and the effective  $\Lambda$  is set by the inverse-area scale of the horizon. The DK framework differs from entropic gravity in providing a *physical mechanism* (low- $\Gamma$  vacuum-fluctuation conversion) rather than a purely thermodynamic argument — but the geometric scaling is the same.

## 6. Testable Predictions

### 6.1 The $H_0$ tension

In standard  $\Lambda$ CDM, the present-day Hubble rate  $H_0$  is a single number characterizing a uniform expansion. The measured tension between CMB-derived (Planck:  $H_0 \approx 67.4$  km/s/Mpc) and local-distance-ladder (SH0ES:  $H_0 \approx 73.0$  km/s/Mpc) values represents a  $\sim 5\sigma$  disagreement that  $\Lambda$ CDM cannot accommodate without modification.

The DK framework predicts that  $H_{\text{local}}$  should depend on the local matter density along the line of sight. The local distance ladder samples regions inside our cosmic neighborhood, including the local void (“KBC void”) in which the Milky Way may sit. CMB measurements average over the entire observable horizon, sampling cosmic-mean density.

If expansion is sourced preferentially in low-density regions, then a local probe biased toward void regions will measure a *higher* effective  $H_0$  than a global probe — qualitatively matching the observed tension.

**Quantitative prediction:**  $H_{\text{local}}/H_{\text{global}}$  should correlate with the ratio of local-mean-density to cosmic-mean-density, with the correlation function determined by the relative weights of  $\Gamma_{\text{env}}$  and  $\Gamma_{\text{grav}}$  suppression in the two density regimes. A first-order estimate predicts a tension of order  $\Delta H_0/H_0 \sim (\rho_{\text{cosmic}} - \rho_{\text{local}})/\rho_{\text{cosmic}}$ ; the observed tension of  $\Delta H_0/H_0 \approx 0.08$  is consistent with the void underdensities of order  $\sim 10\%$  inferred from galaxy surveys.

### 6.2 Void expansion rates

$\Lambda$ CDM predicts that cosmic voids expand at the same Hubble rate as the surrounding universe, modified only by the local gravitational deficit (which slightly *enhances* recession). The DK framework predicts an *additional* contribution: voids should expand faster because real-particle creation is actively occurring within them, sourcing additional expansion not present in dense regions.

**Quantitative prediction:** the void expansion velocity excess over  $\Lambda$ CDM should scale with the void’s emptiness — deeper voids expand faster, with the excess vanishing as the void’s mean density approaches the cosmic mean.

This is in principle measurable through redshift-space distortions in galaxy surveys around well-characterized voids (BOSS, eBOSS, DESI). A positive detection would constitute direct evidence for the proposed mechanism; a null result at the predicted level would falsify it.

### 6.3 The CMB-anisotropy / structure correlation

If the surface-area mechanism is correct, then the rate of expansion should track the surface area of the populated region’s structure at all scales — not just the cosmological horizon. The cosmic web of voids and filaments exposes substantial *internal* surface area, and creation should occur on these internal void surfaces in proportion.

**Quantitative prediction:** small-scale CMB anisotropies should show a slight excess in regions of the sky where galaxy surveys reveal greater total void-surface area along the line of sight. The signal would be small but, in principle, distinguishable from the standard ISW (integrated Sachs–Wolfe) signal because it would correlate with the *surface area* of intervening voids rather than their gravitational potential alone.



## 6.4 The Tolman test as a consistency check

Tolman dimming — the prediction that surface brightness of standard sources falls as  $(1+z)^4$  — is a robust prediction of metric expansion that distinguishes it from non-metric alternatives. The proposed framework *preserves* this prediction: although the substrate is Minkowski, the *effective* metric inside the populated region (§3.4) is FRW, and observers using the effective metric see standard  $(1+z)^4$  dimming. A null result on Tolman dimming would have falsified the framework; the observed  $(1+z)^4$  scaling is consistent with it.

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## 7. Open Questions and Required Derivations

### 7.1 The Goldstone-mode identification

The load-bearing assumption is that excitations of cosmological  $\Phi_{\text{bulk}}$  inherit equation of state  $w = -1$ . This requires showing:

1. That cosmological  $\Phi_{\text{bulk}}$  has a broken-U(1) (or analogous) structure capable of supporting Goldstone modes.
2. That those Goldstone modes are massless (Goldstone’s theorem guarantees this if exact U(1) symmetry is spontaneously broken).
3. That their effective stress-energy tensor takes the pressure-equals-negative-density form  $p = -\rho$  characteristic of  $w = -1$ .

A full derivation would proceed from an effective Lagrangian  $\mathcal{L}[\Phi_{\text{bulk}}]$  written in covariant form on the substrate Minkowski background, with cosmological symmetry breaking producing the order-parameter phase.

### 7.2 The creation-front speed $v_c$

The framework introduces a single dimensional parameter  $v_c$  — the speed at which the boundary of the populated region advances into the substrate. Observationally, the relation  $H_0 = v_c/R_0$  with  $R_0 \approx c/H_0$  implies  $v_c \approx c$ . A first-principles derivation of  $v_c$  from the vacuum-fluctuation lifetime and the DK  $\Gamma$ -suppression structure is needed.

### 7.3 The smooth-universe assumption

Equation (5.2) assumed a spherical populated region with a single boundary. A realistic population would have a fractal-like boundary (cosmic web), with internal void surfaces contributing to the total creation rate. A more complete treatment must compute the total surface area integrated over all scales and check whether the dominant contribution comes from the cosmological horizon or from internal void surfaces.

### 7.4 Reconciliation with primordial nucleosynthesis and CMB

Standard  $\Lambda$ CDM successfully predicts the abundance of light elements (Big Bang nucleosynthesis) and the CMB power spectrum from a hot early universe. The DK framework, if it claims to replace  $\Lambda$  rather than supplement it, must reproduce these successes. The proposal here is that BBN and CMB physics occur *inside* the populated region under the effective FRW metric, and so the standard predictions are preserved by construction — but this needs explicit verification, particularly during the radiation-dominated early epoch when the  $\Gamma$ -suppression mechanism would be deeply in the high- $\Gamma$  regime and thus inactive.

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## 8. Conclusion

The standard FRW framework treats cosmic expansion as a geometric property of spacetime sourced by an axiomatic cosmological constant  $\Lambda$ . This works, but at the cost of three unresolved problems: the 120-order-of-magnitude vacuum-energy discrepancy, the coincidence problem, and the  $H_0$  tension.

The Dirac–Kuramoto framework already provides the conceptual machinery to reframe  $\Lambda$  as emergent: an infinite quantum-field substrate; a finite populated region characterized by bulk coherence  $\Phi_{\text{bulk}}$ ; and explicit Stage-2 relaxation rates  $\Gamma_{\text{env}}(T)$  and  $\Gamma_{\text{grav}}(M)$  that vanish in low-temperature, low-gravitational-density regions. The proposal advanced here is that cosmic expansion is the dynamical growth of the populated region into the substrate, driven by real-particle creation in low- $\Gamma$  regions where bulk coherence cannot suppress vacuum fluctuations in time.

The mechanism naturally produces an expansion rate proportional to the *surface area* of the populated region, yielding  $H \propto 1/R$  and an effective  $\Lambda_{\text{eff}} \propto 1/R^2$  that matches the observed  $\Lambda \sim 1/L_H^2$  without postulation. It predicts three observable distinctions from  $\Lambda$ CDM: a void-correlated  $H_0$  tension, anomalously fast void expansion, and a small CMB-anisotropy correlation with intervening void-surface area. It preserves Tolman  $(1+z)^4$  dimming and the CMB power spectrum through the effective-FRW-metric construction inside the populated region.

The framework is presented here as a working-paper sketch. Its load-bearing assumption — that Goldstone modes of cosmological  $\Phi_{\text{bulk}}$  carry equation of state  $w = -1$  — is identified as the next required derivation. If that identification can be made rigorous, the cosmological-constant problem dissolves:  $\Lambda$  is no longer an arbitrary input but the emergent ground-state property of bulk-coherence incompleteness at horizon scales, with magnitude set dynamically by the size of the universe.

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## Author Contributions

JB conceived the substrate–population distinction, the connection between DK  $\Gamma$ -suppression and vacuum-fluctuation conversion, and the surface-area scaling. Claude Opus 4.7 (Anthropic) contributed mathematical and historical structure, the connection to existing mechanisms (Parker, Schwinger, Hawking, dynamical Casimir, holographic principle), the Goldstone-mode equation-of-state argument, and the manuscript drafting. Per current journal authorship guidelines, LLMs do not satisfy authorship criteria; the human author bears full responsibility for all scientific content. Were such policies not in force, Claude would be listed as co-first author.

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